

Policy Memo | Why Limiting Climate Change Matters for Niger

Advice to the national delegation to the UNFCCC

To: Minister of Environment, Head of Niger's UNFCCC delegation	From: KOUAME Koffi Fidèle, Climate Adviser
Date: 14 October 2026	Subject: Climate risks for Niger and the case for limiting global warming

Summary

Niger faces a sharp rise in dangerous heat. Under current global policies, days of dangerous heat would increase from 31 a year today to 91 by 2100. If warming stays below 2 °C, they would stay near 42. Extreme rainfall would also intensify much more under current policies. Severe drought will affect almost the whole country by 2030 in all scenarios. Niger should therefore support rapid global emission cuts and seek adaptation finance now.

A. Situation

1. Agriculture and livestock generate almost 40 % of GDP and over 80 % of jobs. Both depend on rainfall, and most of this work is done outdoors (World Bank, 2024).
2. Climate shocks are already severe. The 2005 and 2010 droughts caused national food crises. The 2024 floods killed 339 people and affected more than 1.1 million (AFP, 2024).
3. Niger's share of global emissions is negligible. Its exposure therefore depends on the emissions of other countries.

B. Projected impacts

4. **Heat (Table 1, Figure 1).** Dangerous-heat days triple under current policies, from 31 to 91 a year by 2100. Below 2 °C, they stabilise near 42. Above this heat-index threshold, sustained outdoor work exposes people to heat exhaustion and heat stroke. The increase is about twice as large in the populated south as in the north (+44 against +21 days at 2 °C compared with 1 °C of global warming). WASCAL researchers project heat discomfort for over half of West Africa's population at 1.5 to 2 °C (Sylla et al., 2018).
5. **Drought.** The indicator includes evaporation, so it shows how heat dries soils and pastures. About 96 % of the land is affected by 2030 in all scenarios. Limiting warming to 2 °C does not change this. Only a 1.5 °C pathway reduces the share, to 87 % by 2100. Millet yields are expected to fall in most scenarios (Mohamed et al., 2025).
6. **Rainfall.** Extreme rainfall rises by 43 % under current policies by 2100, against 17 % below

2 °C. Even the low end of the current-policies range shows an increase. Flood risk therefore grows in Niamey and the Niger River valley, where people, infrastructure and irrigated rice are concentrated. Extreme Sahelian storms have already tripled since 1982 (Taylor et al., 2017).

Table 1. Projected change in three climate impacts for Niger

Impact (CIE indicator)	Today (2020)	2050, below 2 °C	2050, current policies	2100, below 2 °C	2100, current policies
Dangerous heat (days per year with heat index above 40 °C)	31 [25–38]	45 [33–69]	54 [41–84]	42 [29–72]	91 [59–134]
Severe drought (share of land with at least one month of SPEI below –1.5)	84 % [72–97]	98 % [85–100]	≈100 % [96–100]	98 % [81–100]	100 % [98–100]
Extreme rainfall (annual maximum 5-day rainfall, change from 1995–2014)	+8 % [–7, +29]	+18 % [0, +55]	+24 % [–1, +67]	+17 % [–1, +54]	+43 % [+5, +110]

Source: *Climate Impact Explorer, Niger*; NGFS scenarios (1.6 °C and 2.9 °C of warming in 2100); model median, 5–95 % range in brackets; values after 2060 indicative.

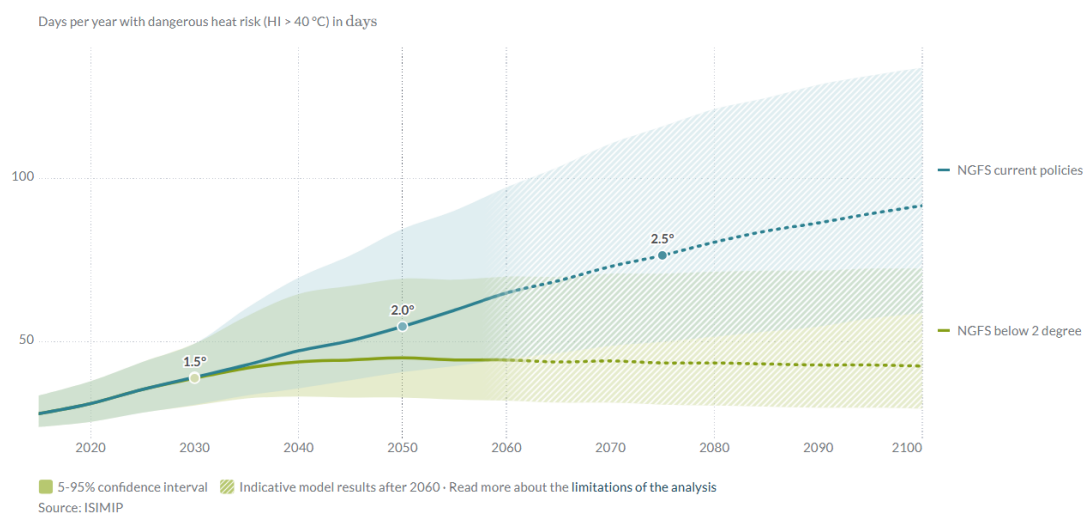


Figure 1. Days of dangerous heat per year in Niger. The two pathways separate after 2030. In 2100, current policies lead to about 91 days a year, against 42 below 2 °C and 31 today. The low end of the current-policies range (59 days) is above the central below-2 °C value. Results after 2060 (hatched) are indicative. Source: Climate Impact Explorer, Climate Analytics.

C. Assessment

7. Limiting warming below 2 °C avoids about 49 days of dangerous heat a year by 2100. It also cuts the rise in extreme rainfall from 43 % to 17 %. A 1.5 °C pathway keeps heat near today's level (32 days a year by 2100).
8. The economic cost is high. Climate change could reduce Niger's annual GDP by 2.2 % in a wet scenario and by 11.9 % in a dry scenario by 2050. These estimates exclude migration and conflict (World Bank, 2022).
9. Drought increases in all scenarios. A dry year reduces harvests and pastures, raises grain prices and forces sales of livestock. This leads to malnutrition, debt and tension between farmers and herders. Adaptation is therefore needed now, whatever the outcome of the negotiations.

D. Recommendations

- (a) Support the 1.5 °C goal. Ask major emitters for national plans consistent with a rapid phase-out of fossil fuels.
- (b) Link Niger's support for any agreement to more adaptation finance, with simple access for Least Developed Countries.
- (c) Seek support from the Loss and Damage Fund for droughts and floods, in coalition with the African Group of Negotiators and the LDC Group.
- (d) At national level, fund drought-tolerant millet and sorghum, water harvesting, early warning systems and the protection of herders' mobility.

E. Limits of the analysis

10. These figures are projections, not forecasts. The direction of change is clear for heat, but its size varies between models (59 to 134 days in 2100 under current policies).
11. The heat index uses an estimate of humidity at the hottest hour. The drought indicator counts a single month of drought, so it saturates. The tool does not model millet or sorghum, so its crop indicators (maize, rice, soy) were not used.
12. These are hazard projections only. Given Niger's high vulnerability, real impacts are likely to be larger.

References

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